

# Neutron Scaling Laws from Numerical Experiments

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**Abstract** Experimental data of neutron yield  $Y_n$  against pinch current  $I_{\text{pinch}}$  is assembled to produce a more global scaling law than available. From the data a mid-range point is obtained to calibrate the neutron production mechanism of the Lee Model code. This code is then used for numerical experiments on a range of focus devices to derive neutron scaling laws. The results are the following:  $Y_n = 2 \times 10^{11} I_{\text{pinch}}^{4.7}$  and  $Y_n = 9 \times 10^9 I_{\text{peak}}^{3.9}$ . It is felt that the scaling law with respect to  $I_{\text{pinch}}$  is rigorously obtained by these numerical experiments when compared with that obtained from measured data, which suffers from inadequacies in the measurements of  $I_{\text{pinch}}$ .

**Keywords** Plasma Focus · Neutron scaling · Pinch current · Focus modeling · Lee Model

## Introduction

A major feature of the plasma focus is its fusion neutron yield. Even a simple trolley mounted 3 kJ device such as the UNU/ICTP PFF routinely produces [1] a yield of  $Y_n = 10^8$  neutrons, operating in deuterium. A big machine such as the PF1000 typically produces  $10^{11}$  neutrons per shot [2]. Moreover since the neutrons are produced in a short pulse of the order of 10 ns, the rate of neutron

production is  $10^{16}$  neutrons/s even for a small machine and can go up to  $10^{20}$  for a large machine.

From a compilation of experimental data over a wide range of energies a scaling law of  $Y_n \sim I_{\text{pinch}}^{3.3}$  was presented by Bernard [3], where  $I_{\text{pinch}}$  is the current flowing through the dense pinch in the focused plasma. Kies [4] presented another compilation showing  $Y_n \sim I_{\text{pinch}}^4$  whilst Herold [5] had results showing  $Y_n \sim I_{\text{pinch}}^{3.2}$ . Gribkov [2] has recently suggested that the experimental data can be interpreted with the power law as high as 5 in particular when dealing with the same device.

One significant uncertainty in compiling such a scaling law is the interpretation of  $I_{\text{pinch}}$ . The current most conveniently measured in most experiments is the total current flowing into the tube (usually measured with a Rogowski coil placed at the collector plate just outside the tube). This total current has a maximum value  $I_{\text{peak}}$ . If one estimates  $I_{\text{pinch}}$  from the total current measurement there are two difficulties: (1) it is difficult to determine the point on the current waveform where the plasma has gone into the pinch phase, and (2) even after estimating this point, it still remains to estimate the fraction of total current that in fact flows into the pinch. One way is to use small magnetic coils to probe the pinch region. For small machines this method is not suitable because of the amount of space available and the small size of the pinch so that the probes inevitably interfere with the pinching current sheet. For large machines, results have been obtained [5] but with large errors quoted as 20%. Moreover the shot-to-shot variability of focus performance means that the final presentation of results relies greatly on how the particular research group chooses to present the results. For example the yield may be presented as a range, with some shots considered not representative discarded, and perhaps the biggest values of observed yield also presented. It is quite

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remarkable that despite all these difficulties there is a consensus of opinion that the index in this power scaling law has the value in the range of 3–5.

### Compilation of Experimental Results

In this paper we have combined the laboratory data that we have [1–7], which includes recent results from some smaller machines e.g. Soto's [6] PF400 and the large [2] PF1000 as well as a high performance repetitive device [7], the NX2. This gives a good fit of  $Y_n = 9 \times 10^{10} I_{\text{pinch}}^{3.8}$ . The main reason for this compilation of experimental results is to provide a calibration point for setting the neutron yield mechanism of the Lee Model code, described below. A calibration point is chosen at around the middle of the current range at  $I_{\text{pinch}} = 0.5$  MA,  $Y_n = 6 \times 10^9$  neutrons. This point is close to the PF1000's machine parameters with properly adjusted dimensions if it could be fired at 13.5 kV.

The results of the compilation are shown in Fig. 1.

### The Model used for the Numerical Experiments

The Lee Model has been widely used to simulate axial and radial phase dynamics, temperatures and thermodynamic properties and radiation yields. To realistically simulate any plasma focus all that is needed is a measured current trace of that plasma focus. Recently the model code [8] has been extended to include a phenomenological beam-target

mechanism based partially on that proposed by Gribkov [2].

The main mechanism producing the neutrons is a beam of fast deuteron ions interacting with the hot dense plasma of the focus pinch column. The fast ion beam is produced by diode action in a thin layer close to the anode with plasma disruptions generating the necessary high voltages. This mechanism, described in some details in a recent paper [9], results in the following expression used for the model code:

$$Y_{b-t} = \text{calibration constant} \times I_{\text{pinch}}^2 z_p^2 (\ln(b/r_p)) \sigma / V_{\text{max}}^{0.5}$$

where  $I_{\text{pinch}}$  is the current at the start of the slow compression phase,  $r_p$  and  $z_p$  are the pinch radius and pinch length at the end of the slow compression phase,  $V_{\text{max}}$  is the maximum value attained by the inductively induced voltage and  $\sigma$  is the D–D fusion cross section ( $n$  branch) [10] corresponding to the beam ion energy. The D–D cross section  $\sigma$  is obtained by using beam energy equal to three times  $V_{\text{max}}$ , to conform to experimental observations.

### Scaling Laws Derived from the Numerical Experiments

This paper applies the code to several machines including the PF400, UNU/ICTP PFF, the NX2 and Poseidon. The PF1000 which has a current curve published at 27 kV and  $Y_n$  published at 35 kV provided an important point. Moreover using parameters for the PF1000 established at 27 and 35 kV, additional points were taken at different voltages ranging from 13.5 kV upwards to 40 kV.

These machines were chosen because each has a published current trace and hence the current curve computed by the model code could be fitted to the measured current trace. Once this fitting is done our experience is that the other computed properties including dynamics, energy distributions and radiation are all realistic. This gives confidence that the computed  $Y_n$  for each case is also realistic. Moreover since each chosen machine also has measured  $Y_n$  corresponding to the current trace, the computed  $Y_n$  could also be compared with the measured to ensure that the computed results are not incompatible with the measured values.

The results are shown in Table 1 and Fig. 2.

In Table 1, corresponding to each laboratory device, the operating voltage  $V_o$  and pressure  $P_o$  are typical of the device, as is the capacitance  $C_o$ . It was found that the static inductance  $L_o$  usually needed to be adjusted from the value provided by the laboratory. This is because the value provided could be for short circuit conditions, or an estimate including the input flanges and hence that value may not be sufficiently close to  $L_o$ . The dimensions  $b$

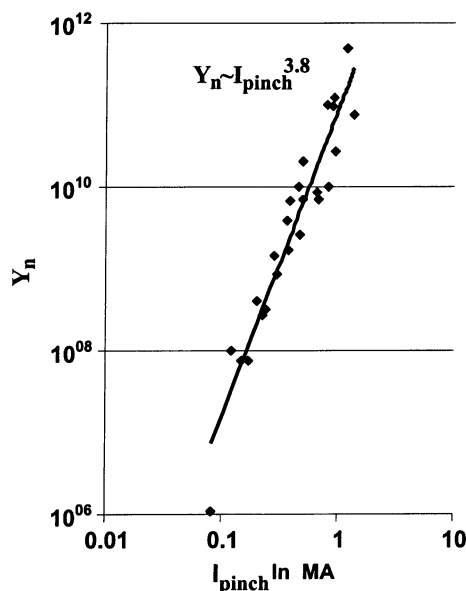
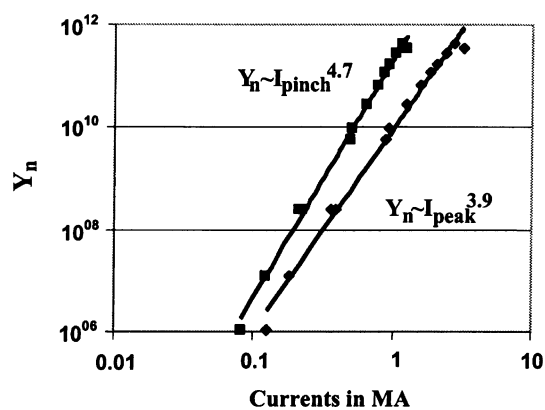


Fig. 1  $Y_n$  scaling with  $I_{\text{pinch}}$  from laboratory data

**Table 1** Computed values of  $I_{\text{peak}}$ ,  $I_{\text{pinch}}$  and  $Y_n$  for a range of plasma focus machines

| Machine     | $V_o$ (kV) | $P_o$ (torr) | $L_o$ (nH) | $C_o$ ( $\mu$ F) | $b$ (cm) | $a$ (cm) | $Z_o$ (cm) | $I_{\text{peak}}$ (MA) | $I_{\text{pinch}}$ (MA) | $S$ | $Y_n$                | $k_{\text{min}}$ | $I_{\text{pinch}}/I_{\text{peak}}$ |
|-------------|------------|--------------|------------|------------------|----------|----------|------------|------------------------|-------------------------|-----|----------------------|------------------|------------------------------------|
| PF400       | 28         | 6.6          | 40         | 0.95             | 1.55     | 0.60     | 1.7        | 0.126                  | 0.082                   | 82  | $1.1 \times 10^{06}$ | 0.14             | 0.65                               |
| UNU         | 15         | 4            | 110        | 30               | 3.2      | 0.95     | 16         | 0.182                  | 0.123                   | 96  | $1.2 \times 10^{07}$ | 0.14             | 0.68                               |
| NX2 T       | 15         | 5            | 20         | 28               | 5        | 2        | 7          | 0.386                  | 0.225                   | 86  | $2.5 \times 10^{08}$ | 0.16             | 0.58                               |
| Calibration | 16         | 5            | 24         | 308              | 7        | 4        | 30         | 0.889                  | 0.496                   | 99  | $5.6 \times 10^{09}$ | 0.17             | 0.56                               |
| NX2 T-2     | 12.5       | 10.6         | 19         | 28               | 3.8      | 1.55     | 4          | 0.357                  | 0.211                   | 71  | $2.4 \times 10^{08}$ | 0.16             | 0.59                               |
| PF1000      | 13.5       | 3.5          | 33         | 1332             | 8.00     | 5.78     | 60         | 0.924                  | 0.507                   | 89  | $9.6 \times 10^{09}$ | 0.17             | 0.55                               |
|             | 18         | 3.5          | 33         | 1332             | 10.67    | 7.70     | 60         | 1.231                  | 0.636                   | 89  | $2.9 \times 10^{10}$ | 0.18             | 0.52                               |
|             | 23         | 3.5          | 33         | 1332             | 13.63    | 9.84     | 60         | 1.574                  | 0.766                   | 89  | $6.8 \times 10^{10}$ | 0.19             | 0.49                               |
|             | 27         | 3.5          | 33         | 1332             | 16       | 11.60    | 60         | 1.847                  | 0.862                   | 89  | $1.2 \times 10^{11}$ | 0.19             | 0.47                               |
|             | 30         | 3.5          | 33         | 1332             | 17.77    | 12.80    | 60         | 2.049                  | 0.929                   | 89  | $1.6 \times 10^{11}$ | 0.20             | 0.45                               |
|             | 35         | 3.5          | 33         | 1332             | 20.74    | 15.00    | 60         | 2.399                  | 1.037                   | 89  | $2.7 \times 10^{11}$ | 0.20             | 0.43                               |
| Poseidon    | 40         | 3.5          | 33         | 1332             | 23.70    | 17.10    | 60         | 2.736                  | 1.137                   | 89  | $4.1 \times 10^{11}$ | 0.21             | 0.42                               |
|             | 60         | 3.8          | 18         | 156              | 9.50     | 6.55     | 30         | 3.200                  | 1.260                   | 251 | $3.3 \times 10^{11}$ | 0.20             | 0.39                               |

**Fig. 2**  $Y_n$  scaling with  $I_{\text{pinch}}$  and  $I_{\text{peak}}$  from numerical experiments

(outer radius),  $a$  (anode radius) and  $z_o$  (anode length) are also the typical dimensions for the specific device. The speed factor [11]  $S$  is also included. All devices except Poseidon have typical  $S$  values. Poseidon is the exceptional high speed device in this respect. The minimum pinch radius is also tabulated as  $k_{\text{min}} = r_p/a$ . It is noted that this parameter increases from 0.14 for the smaller machines towards 0.2 for the biggest machines. The ratio  $I_{\text{pinch}}/I_{\text{peak}}$  is also tabulated showing a trend of decreasing from 0.65 for small machines to 0.4 for the biggest machines.

The resultant data with improved optimization yield more up to date scaling laws:  $Y_n \sim I_{\text{pinch}}^{4.7}$  and  $Y_n \sim I_{\text{peak}}^{3.9}$ . It is necessary to emphasize again that the  $I_{\text{pinch}}$  may be considered to be computed rigorously especially for those cases where an experimental current curve is available. Once the computed current curve is fitted accurately to the experimental current curve, the resultant pinch position is pinpointed as well as the fraction of current going into the pinch. This is in contrast to the laboratory data where  $I_{\text{pinch}}$  is usually only estimated and if measured is subject to large

errors. A study of the data suggests that in most cases  $I_{\text{pinch}}$  is overestimated by experimentalists. With all these considerations it would appear that the scaling laws arising from the code are not inconsistent with experimental observations and may complement the more conventionally compiled scaling laws to provide comprehensive database for experiments.

## Conclusion

Neutron scaling laws have been derived from computation using the Lee Model code. These are:  $Y_n \sim I_{\text{pinch}}^{4.7}$  and  $Y_n \sim I_{\text{peak}}^{3.9}$ . In these numerical experiments  $I_{\text{pinch}}$  is rigorously computed whereas in compilation of laboratory results  $I_{\text{pinch}}$  is usually just guessed at or at best estimated. These numerically derived scaling laws are not inconsistent with compilation from laboratory experiments. The numerically derived scaling law against  $I_{\text{pinch}}$  has an index of 4.7 which is higher than the usually accepted scaling law with index of 3.2 to 4. The indications are that the numerically derived scaling laws being more rigorous and consistent in derivation may actually be more realistic and more reliable for use in interpreting, designing or planning experiments.

## References

1. S. Lee, T.Y. Tou, S.P. Moo, M.A. Elissa, A.V. Gholap, K.H. Kwek, S. Mulyodrono, A.J. Smith, Suryadi, W. Usala, M. Zak-aulah, Am. J. Phys. **56**, 62 (1988)
2. V.A. Gribkov, A. Banaszak, B. Bienkowska, A.V. Dubrovsky, I. Ivanova-Stanik, L. Jakubowski, L. Karpinski, R.A. Miklaszewski, M. Paduch, M.J. Sadowski, M. Scholz, A. Szydlowski, K. Tomaszewski, J. Phys. D: Appl. Phys. **40**, 3592 (2007)

3. A. Bernard, A. Coudeville, J.P. Garconnet, A. Jolas, J. de Mascureau, C. Nazet, J. Phys. Colloque C1, **39**(5), C1 (1978)
4. W. Kies, in *Laser and Plasma Technology*, ed. by S. Lee et al. Procs of Second Tropical College (World Scientific, Singapore, 1988) ISBN 9971-50-767-6, pp. 86–137
5. H. Herold, in *Laser and Plasma Technology*, ed. by C.S. Wong et al. Procs of Third Tropical College (World Scientific, Singapore, 1990) ISBN 981-02-0168-0, pp. 21–45
6. L. Soto, P. Silva, J. Moreno, G. Silvester, M. Zambra, C. Pavez, L. Altamirano, H. Bruzzone, M. Barbaglia, Y. Sidelnikov, W. Kies Brazilian, J. Phys. **34**, 1814 (2004)
7. A. Patran, R.S. Rawat, J.M. Koh, S.V. Springham, T.L. Tan, P. Lee, S. Lee, in *31st EPS Conference on Plasma Phys. London, 2004 ECA*, vol. 28G, P–4.213 (2004)
8. S. Lee, Radiative Dense Plasma Focus Computation Package: RADPF, in <http://www.intimal.edu.my/school/fas/UFLF/>. Accessed 24 December 2007
9. S. Lee, S.H. Saw, Pinch current limitation effect in plasma focus. Appl. Phys. Lett. **92**, 021503 (2008)
10. J.D. Huba, Plasma Formulary pg44, (2006), [http://wwwppd.nrl.navy.mil/nrlformulary/NRL\\_FORMULARY\\_07.pdf](http://wwwppd.nrl.navy.mil/nrlformulary/NRL_FORMULARY_07.pdf). Accessed 15 October 2007
11. S. Lee, A. Serban, IEEE Trans. Plasma Sci. **24**, 1101–1105 (1996)